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SHORT-FLAME/QUICK-QUENCH: A UNIQUE ULTRALOW EMISSIONS COMBUSTION CONCEPT FOR GAS TURBINE COMBUSTORS

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This paper describes the emission characteristics of model combustors employing a unique low-emissions gas turbine combustion concept named *short-flame/quick-quench combustion.* Its potential for ultralow NO_x emissions and complete combustion has previously been successfully demonstrated by testing in an engine. This concept combines lean premixed combustion in a short reaction zone generated by many small flames stabilized on a perforated plate flame holder, and quick quenching of NO_x formation by injecting dilution air immediately downstream of the reaction zone. Multiple tubes penetrate through the flame holder and extend into the combustion chamber, and dilution air is injected from holes on the tubes at an axial position where major combustion reactions have been completed, to achieve both ultralow NO_x emissions and complete combustion. Measurements of gaseous emissions from the model combustors were made for natural gas at combustor inlet air temperatures ranging from 300 to 900 °C and pressures of 0.1 to 0.7 MPa. Modulation of combustion and dilution air split resulted in NO_x emissions of 3–4 ppm (corrected at 15% O_2) or 0.3–0.4 $EINO_x$ (g NO_2/kg fuel) over the range of overall equivalence ratios required in simple- and regenerative-cycle gas turbine operations. The NO_x concentrations in parts per million correlated well by the Arrhenius expression with the adiabatic flame temperature in the reaction zone, T_p, over a wide range of pressure, temperature, and air split between combustion and dilution. It is shown that $NO_x \propto \exp(-(50 \pm 2) \times 10^3/RT_p)$, where R is the universal gas constant in cal/mol K.

Introduction

Lean premixed combustion has the potential to reduce NO_x emissions from various combustion devices to levels 2 orders of magnitude lower than those of diffusion flame combustion. One of the inherent problems associated with the application of the low-NO_x combustion concept to gas turbine combustors is the difficulty in the trade-off between CO and NO_x emissions. The magnitude of NO_x reduction achieved just by decreasing the mixture equivalence ratio is limited by an unacceptable increase of CO emissions at the lean end of operations. Because of the three-dimensional, complicated flow fields in the combustor, the residence time in the reaction zone varies widely in space: it is too short for CO oxidation in some regions, while it is too long for suppressing NO_x formation in other regions, which results in the difficult trade-off between CO and NO_x emissions.

In order to achieve single-digit parts per million NO_x emissions from gas-fueled stationary gas turbines and a better NO_x-CO trade-off, a unique combustion concept, short-flame/quick-quench low-NO_x combustion, has been developed. In the preliminarily stage of the development, NO_x formation and CO destruction were investigated in the postflame region of a quasi-one-dimensional premixed reaction zone consisting of many tiny flames stabilized on a ceramic honeycomb flame holder placed in a ceramic flame tube. The CO concentration decreased steeply to an acceptable level at a very short distance from the flame holder, while the NO_x concentration continued to increase with axial distance, especially when the burned gas temperature was high. It was found that the introduction of air in the postflame region close to the end of the reaction zone resulted in complete quenching of NO_x formation, as well as complete combustion. Proper injection of dilution air into the postflame hot burned gas in a gas turbine combustor, if possible, would make it easier to reduce CO and NO_x emissions simultaneously. The spatial heat release rate in the premixed reaction zone of this new combustion concept is much higher than that of conventional combustors.

In-engine testing of the concept was successfully conducted by using a 300 kW simple-cycle gas turbine with a pressure ratio of 8 to demonstrate 5 ppm (15% O_2) NO_x emissions near and at full load [1]. This paper describes the emission characteristics of model combustors tested, before the in-engine testing, over a range of representative operating conditions, including inlet air temperature, pressure, and equivalence ratios, of simple- and regenerative-cycle gas turbines.

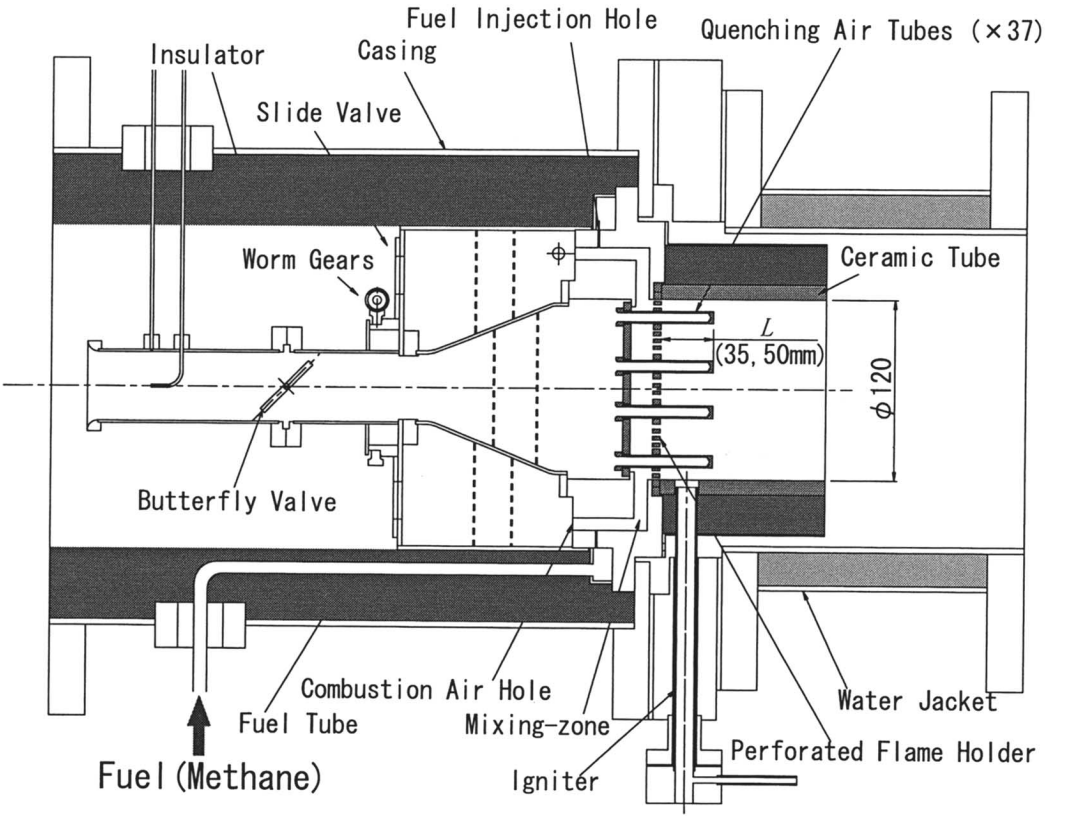

FIG. 1. Schematic drawing of model combustor used for elevated pressure tests.

Low-NO_x Combustion Concept

In the short-flame/quick-quench combustion concept, lean premixed combustion and quick quenching of NO formation by injection of dilution air in the postflame burned gas are combined to achieve both ultralow NO_x emissions and complete combustion. As is seen from the schematic illustration of the model combustor in Fig. 1, a perforated plate flame holder was used to stabilize a large number of small premixed flames so that these flames, as a whole, made a short reaction zone. Multiple cylindrical tubes with closed ends penetrated through the perforated flame holder, extending into the combustion chamber, and dilution air was injected into the burned gas to quench further NO_x formation as well as to limit the residence time in the higher-temperature reaction zone. The mixture equivalence ratio was selected depending on the combustor inlet conditions so that combustion could be completed in the reaction zone. The gas temperature after dilution was controlled in the preferred range of 1200–1400 °C to promote continuing CO oxidation downstream without producing NO_x. The residence time in the reaction zone could be optimized by changing the length of the dilution air tubes.

Burner and Combustor Test Rig

A part of the present research was conducted in the Ceramic Gas Turbine Development Program sponsored by the Ministry of International Trade and Industry Japan, where technologies for 300 kW regenerative-cycle gas turbines with a thermal efficiency greater than 42% were being developed. The model combustor shown in Fig. 1 was used in high-pressure tests, while another model combustor of a similar configuration was used in the atmospheric-pressure tests. The flame holders and quenching air tubes of these model combustors were made of reaction-bonded silicon nitride. Fig. 2 shows a photo of the burner assembly prepared for the elevated pressure tests; it was composed of a flame holder 120 mm in diameter and 37 quenching air tubes with 8 mm outer diameter and 5.8 mm inner diameter. The flame holder had 300 holes 3.4 mm in diameter and 37 holes 9 mm in diameter through which quenching air tubes penetrated. The blockage was 65.9%. Each quenching air tube had a total of eight dilution air injection holes 2 mm in diameter in two rows of 4 and 7 mm from the tip. The combustor for atmospheric pressure tests consisted of a 110 mm diameter flame holder with 282 holes 3.2 mm in diameter

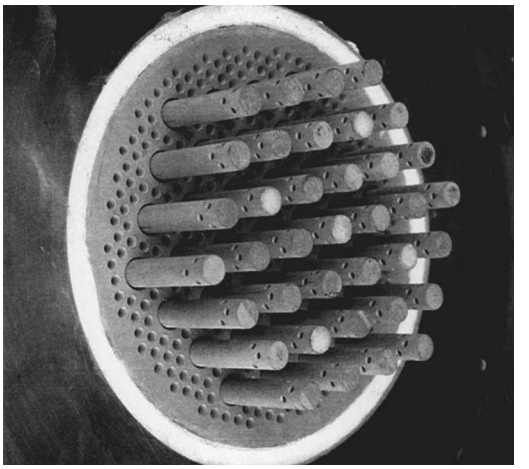

FIG. 2. Photo of burner assembly for combustor used at elevated pressurs, consisting of a perforated flame holder and quenching air tubes made of silicon nitride.

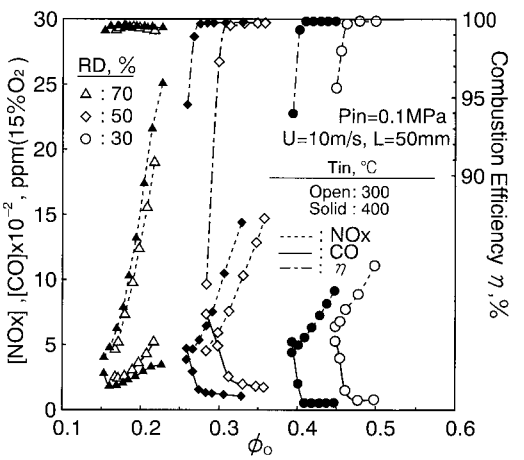

FIG. 3. NO_x and CO concentrations and combustion efficiency as a function of overall equivalence ratio ϕ_o for reaction zone length $L = 50$ mm at atmospheric pressure.

(63.1% blockage) and 37 quenching air tubes with 8 mm outer diameter and 7 mm inner diameter.

Both combustors had independent flow passages and flow control valves for modulating combustion air and quenching air flow rates. A cylindrical straight flame tube 120 mm in diameter was used at elevated pressures, while a 110 mm diameter flame tube with a converging exit 80 mm in diameter was used at atmospheric pressure.

The natural gas fuel (98% methane) was supplied from pressurized cylinders to the burner through a mass flow controller and injected from holes connected to a common fuel manifold in the flange plate to facilitate mixture preparation before the plenum chamber. Measurements of pitot pressure and fuel/air ratio of the mixture jets from the holes on the flame holder showed that the velocity and fuel/air ratio distributions were uniform across the surface. Electric heaters were used to supply non-vitiated air at temperatures ranging from 300 to 900 °C.

Experimental Procedure and Emissions Measurements

Gaseous emissions were measured at pressures between 1 and 7 atm in two inlet air temperature ranges, 300–400 °C and 700–900 °C. The lower temperature range covers typical inlet air temperatures of simple-cycle stationary gas turbines, and the other one represents those of regenerative gas turbine applications.

Exhaust gas was sampled using an X-shaped, water-cooled stainless steel gas sampling probe placed at 90 mm from the surface of the flame holder in the atmospheric-pressure tests, and at 85 mm in the elevated-pressure tests. Heated water was used for gas sampling probe cooling to prevent water vapor from condensing in the probes. Otherwise, some NO_2 may be lost into the condensed liquid phase. The gas temperature detected by a thin thermocouple placed in the gas passage in the probe was monitored throughout the measurements. Standard gas analysis procedures were employed to determine the composition of burned gas: NO was measured by chemiluminescence, O_2 was measured by paramagnetic pressure difference, and CO and CO_2 were measured by non-dispersed infrared absorption. The efficiency of the NO_2-NO converter was verified to be appropriate.

Results

Atmospheric Pressure Data

Figure 3 shows CO and NO_x concentrations (corrected to 15% O_2) and combustion efficiency for a quenching air tube length $L = 50$ mm, at inlet air temperatures, $T_{in} = 300$ and 400 °C. The overall equivalence ratio, ϕ_o; is estimated from the carbon balance of gas compositions determined by the gas analysis. The reference velocity, U, is defined as the ratio of the volumetric flow rate of air at inlet pressure and temperature conditions divided by the cross-sectional area of the flame tube. The parameter RD is used to denote the ratio of mass of quenching air to the total air flow rates. Data for unburned hydrocarbons emissions are not shown in this report since their concentrations were in several parts per million. General trends of emissions and combustion efficiency at these two inlet temperatures are consistent. NO_x emissions increase more steeply with

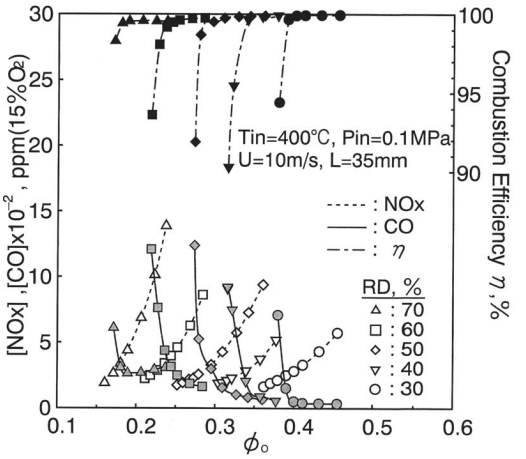

FIG. 4. NO_x and CO concentrations and combustion efficiency as a function of overall equivalence ratio ϕ_o for $L = 35$ mm at atmospheric pressure.

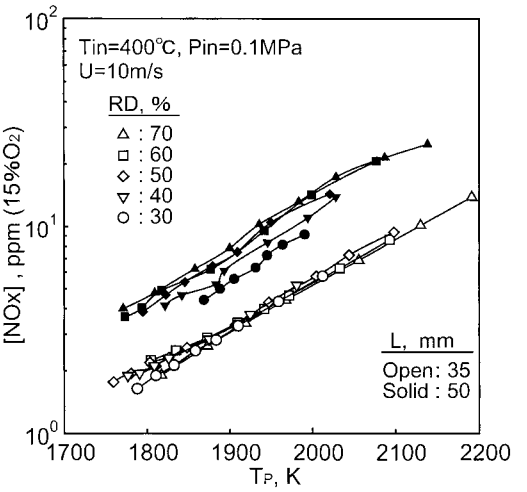

FIG. 5. Effects of reaction zone length and air split on NO_x emissions at atmospheric pressure.

the overall equivalence ratio at larger *RD*. Complete combustion or very low CO emission is achieved when the overall equivalence ratio is greater than a certain critical value for each value of *RD*. The change in the dependence of all emissions on overall equivalence ratio with *RD* is, as a matter of fact, explained not by the overall equivalence ratio but by the equivalence ratio of the mixtures.

Emissions measurements were also made for $L = 35$ mm to investigate the effects of the residence time in the reaction zone on the emissions levels. Typical data for this shorter residence time are shown in Fig. 4. The variation in emissions with overall equivalence ratio exhibits qualitatively the

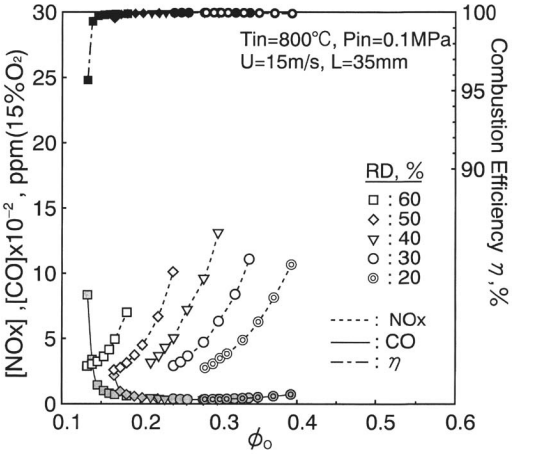

FIG. 6. NO_x and CO emissions and combustion efficiency as a function of overall equivalence ratio ϕ_o for $L = 35$ mm at 800 °C inlet air temperature and atmospheric pressure.

same trends as observed for $L = 50$ mm (Fig. 3). The quantitative difference in NO_x emissions for the two reaction zone lengths is more clearly shown in Fig. 5, where NO_x concentrations (15% O_2) are plotted as a function of reaction zone gas temperature, T_p, the adiabatic gas temperature calculated from the inlet air temperature and equivalence ratio of the mixture. Each set of data points for the same *RD* and *L* are well represented by a straight line, $[NO_x] \propto \exp(5 \times 10^{-3}\, T_p)$, over a wide range of T_p, showing that the NO_x, emissions index ($EINO_x$) is actually an exponentially increasing function of reaction temperature. Higher NO_x emissions at increased *RD* for $L = 50$ mm are due to the longer residence time associated with the reduced mixture flow rate in the reaction zone. The observation that the effects of *RD* on NO_x emissions are smaller for $L = 35$ mm is attributable to the fact that the postflame NO_x formation is relatively less and NO_x produced in the flame front contributes to the major part of the measured NO_x emissions. A decrease of *L* from 50 mm to 35 mm resulted in reductions of NO_x emissions ranging from about 1/2 to 2/3, depending on *RD*. Judging from the low CO emissions shown in Fig. 4, $L = 35$ mm is a better choice for the NO_x-CO emissions trade-off at an inlet air temperature of 400 °C and atmospheric pressure. The proposed concept is advantageous in achieving low emissions: single-digit parts per million NO_x emissions can be achieved at T_p as high as 2000 K without any increase in CO emission.

The concept is also very effective for high air temperature applications in regenerative-cycle gas turbines. Fig. 6 shows the results for $L = 35$ mm at 800 °C inlet air temperature. Minimum NO_x emissions levels achieved at complete combustion are

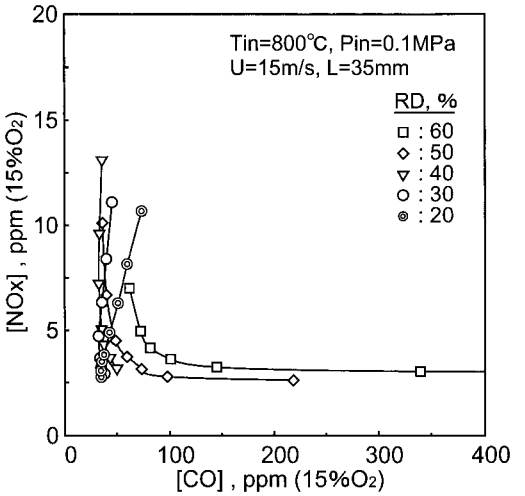

FIG. 7. NO_x emissions versus CO emission for different air splits at 800 °C inlet air temperature and atmospheric pressure.

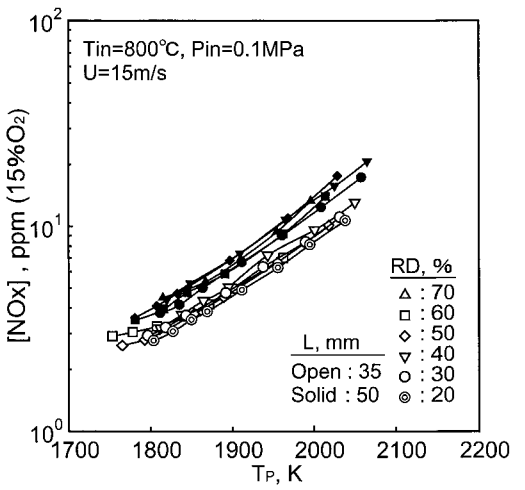

FIG. 8. Effects of reaction zone length and air split on NO_x emissions at 800 °C inlet air temperature and atmospheric pressure.

3 ppm, which do not exceed the levels obtained in the 300–400 °C inlet air temperature range (Figs. 3 and 4). Additionally, very low CO emissions (i.e., complete combustion) were achieved over a wide range of overall equivalence ratios, in spite of an increase in U from 10 m/s to 15 m/s. A correlation of NO_x with CO emissions is shown in Fig. 7. Both NO_x and CO emissions can be reduced simultaneously with decreasing overall equivalence ratio at smaller values of RD, and the minimum NO level is as low as 2.5 ppm for all values of RD.

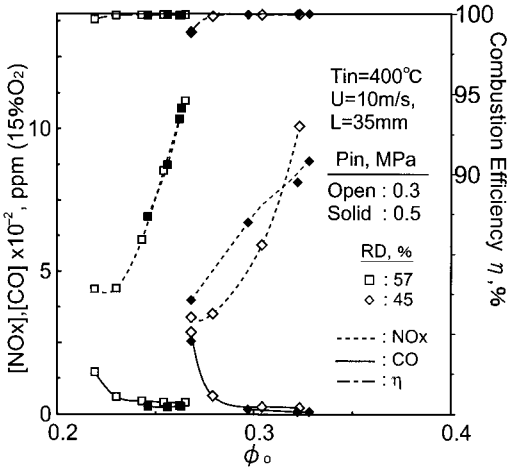

FIG. 9. NO_x and CO emissions and combustion efficiency as a function of overall equivalence ratio ϕ_o for L = 35 mm at 400 °C inlet air temperature and elevated pressures.

The NO_x concentrations (15% O_2) at 800 °C inlet air temperature are shown in Fig. 8 as a function of T_p for different quenching tube lengths. The levels of NO_x at 900 °C inlet air temperature, not shown here, were found to be higher by a small degree than those measured at 800 °C inlet temperature at corresponding conditions. A comparison with the data at 400 °C inlet air temperature, shown in Fig. 5, shows that the difference in NO_x concentration becomes smaller at 800 °C inlet air temperature. An increase in reference velocity from 10 m/s to 15 m/s, however, had a fairly small effect on $EINO_x$ (not shown here). The levels of NO_x emissions in the 700–900 °C inlet air temperature range are also well correlated by T_p and agree with those in the 300–400 °C inlet air temperature range. These results show that the concept can be very effective in achieving ultralow NO_x emissions and complete combustion at inlet air temperatures as high as 900 °C.

Elevated Pressure Tests

Following the atmospheric emissions measurements, elevated pressure emissions measurements were conducted at pressures (P_{in}) of 0.3, 0.5, and 0.7 MPa and at inlet air temperatures of 400, 700, 800, and 900 °C for L = 35 mm. Fig. 9 shows CO and NO_x emissions and combustion efficiency measured at pressures of 0.3 and 0.5 MPa and an inlet air temperature of 400 °C. As with the data at atmospheric pressure (Fig. 4), CO levels became lower and hence high combustion efficiencies were achieved over a wider range of equivalence ratios. NO_x emissions smaller than 10 ppm were measured at elevated

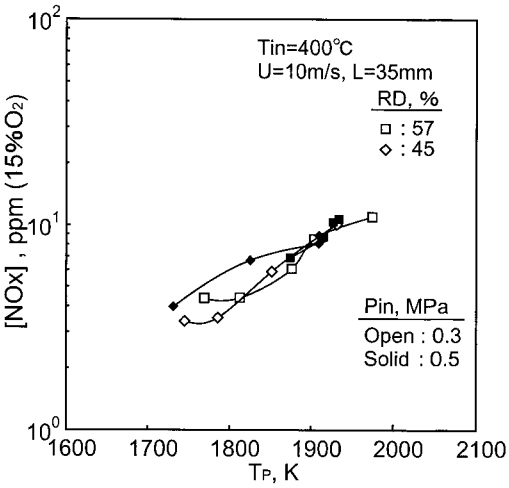

FIG. 10. Effects of pressure and air split on NO_x emissions at 800 °C inlet air temperature.

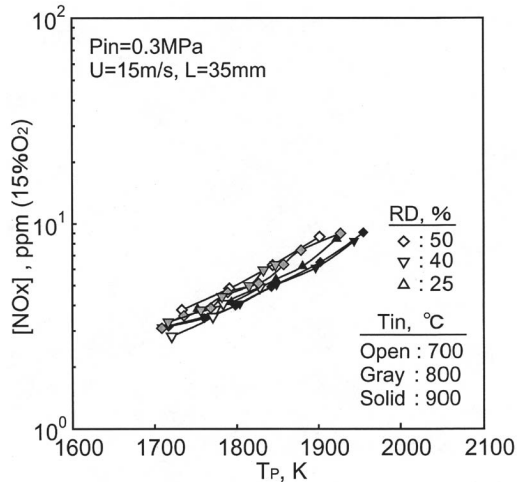

FIG. 12. Effects of inlet air temperature and air split on NO_x emissions at 0.3 MPa pressure.

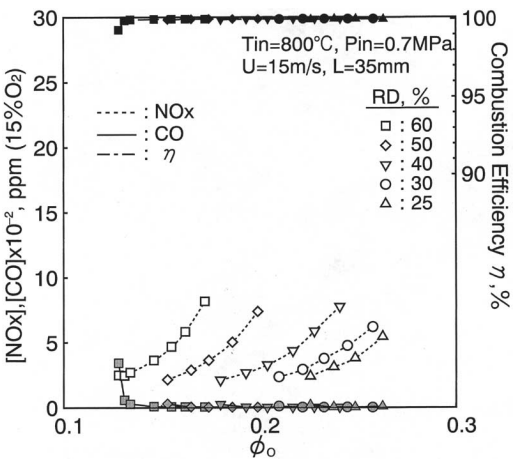

FIG. 11. NO_x and CO emissions and combustion efficiency as a function of overall equivalence ratio ϕ_o for L = 35 mm at 800 °C inlet air temperature and 0.7 MPa pressure.

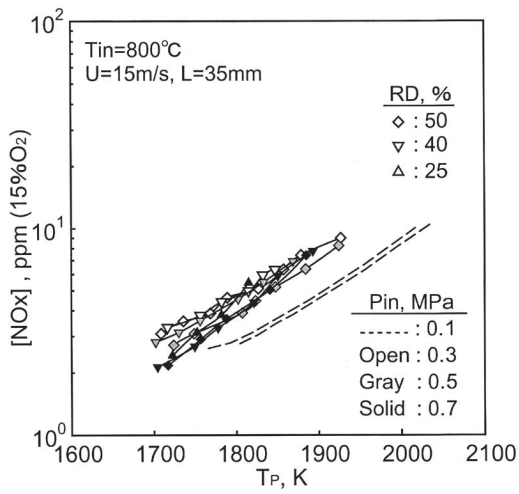

FIG. 13. Effects of pressure and air split on NO_x emissions at 800 °C inlet air temperature.

pressures while maintaining combustion efficiency greater than 99.5%. The NO_x emissions data are plotted as a function of T_p in Fig. 10.

Figure 11 shows CO and NO_x concentrations and combustion efficiency measured at 800 °C inlet air temperature, showing a typical trend for these emissions and combustion efficiency in the higher inlet temperature region. The general trend of the dependence of NO_x emissions on the overall equivalence ratio is similar to that in the lower inlet air temperature range. Qualitatively, CO emissions are very low and combustion is complete even at equivalence ratios as small as 0.13. It is worth mentioning that a better NO_x and CO trade-off can be more easily achieved at higher inlet air temperatures. The minimum NO_x emissions under complete combustion is about 3 ppm at 800 °C inlet air temperature. NO_x concentrations measured for different inlet air temperatures of 700 to 900 °C at 0.3 MPa are correlated with T_p in Fig. 12. The comparison suggests that NO_x emissions are generally higher at higher inlet air temperatures.

The effects of pressure on $EINO_x$ are shown in Fig. 13. The effect of pressure on NO_x emissions is not simple. It is not possible to derive a unique pressure index for expressing pressure effects on $EINO_x$ ($EINO_x$; $\propto p^n$) over the range of T_p tested, even

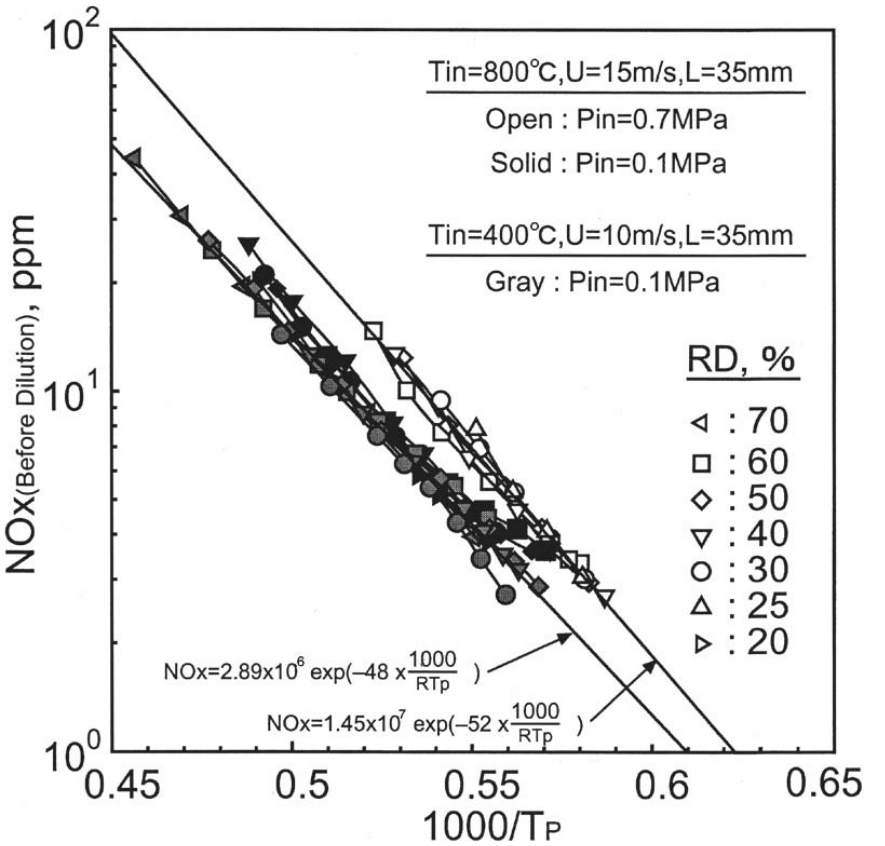

FIG. 14. Arrehenius plots of actual NO_x concentration in the reaction zone at different operating conditions.

though NO_x levels at pressures of 0.3–0.7 MPa were quite a lot higher than those at atmospheric pressure. This trend generally agrees with the chemical kinetic computations for NO_x formation in a model combustor composed of a well-stirred reactor, a one-dimensional quenching air injection zone, and a plug flow reactor [2]. The slope is the steepest at 0.7 MPa. The Arrhenius plot of actual NO_x concentrations in the reaction zone, estimated from the measured NO_x concentration in the sampled gas is shown in Fig. 14, where NO_x concentration in parts per million is plotted against the inverse of the reaction temperature. Data for the same inlet air temperatures are well correlated by single lines which are expressed by $NO_x \propto \exp(-50 \pm 2) \times 10^3/RT_p)$. Xie and Hayashi [3] investigated NO_x formation in an impinging jet methane combustor with minimum heat losses and reported a value of -63×10^3 cal/mol for overall activation energy, E, for $T_p > 1700$ K. The reaction $N_2 + O \rightarrow NO + N$, the rate controlling reaction for NO_x formation by the Zeldovich mechanism, has an activation energy of 75×10^3 cal/mol [4]. The similarity of the values for overall activation energy shows that thermal NO_x is the predominant mechanism of NO_x formation in the present low-emissions combustion concept, short-flame/quick-quench combustion.

Conclusions

A unique low-NO_x combustion concept for gas-fueled gas turbine combustor applications was successfully tested, demonstrating ultralow NO_x emissions of a few parts per million without any compromise of combustion efficiency or CO emission in the temperature and pressure range of simple and regenerative gas turbines. Higher inlet air temperature was favorable for simultaneous achievement of ultraflow NO_x emissions and complete combustion. The overall activation energy for NO_x in parts per million plotted as a function of $1/T_p$ was estimated as $(50 \pm 2) \times 10^3$ cal/mol.

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4. Miller, J. A., and Bowman, C. T., *Prog. Energy Combust. Sci.* 15:287–338 (1989).

COMMENTS

C. H. Priddin, Rolls Royce plc., UK. Do you have any evidence how long the projecting dilution tubes will survive in the gas turbine environment?

Author's Reply. The tubes of the combustor for on-engine testing were made of Inconel, whereas those of the combustor used for the study presented here were made of silicon nitride. We have no data to estimate how long these ceramic tubes survive in the actual gas turbine environment. An inspection of the color of the metal tubes after on-engine testing shows that the metal tubes can survive for a reasonable period in the gas turbine environment, 300 K inlet air temperature and 8 bar pressure.

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Jerald A. Cole, GE Energy & Environmental Research, USA. As the fuel input to the burner is varied over a normal engine turndown cycle, the dilution ratio, R_o, must be continually adjusted to maintain the optimal NO_x performance. It is not clear, however, that optimal NO_x is achieved through a constant total air flow rate, or whether better performance would be achieved if the air flow rate changed with engine load. Could the authors comment on whether it is necessary to vary the total air flow rate to maintain optimal combustor performance throughout turndown, and what impact that might have on engine performance efficiency?

Author's Reply. The dilution ratio has to be continually adjusted to maintain the mixture equivalence ratio to achieve the optimum performance over an engine cycle. It is not necessary to keep the total air flow rate in engine applications, although the experiments were conducted at constant air flow rates. If the total air flow rate to the combustor increases with power or fuel flow rate, the range of modulation of the air split between combustion and dilution can be smaller as compared with a constant air flow rate case.